David Kofi Saah

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EDUCATION

University of Nevada, Reno (UNR) — **Reno, NV, USA**
Ph.D. Statistics and Data Science, CGPA 3.948/4.00 — *August 2025*
Dissertation: Flexible Statistical Models for Imprecise and Uncertain Data: A New Generalization Approach
Advisor: Tomasz J. Kozubowski, Ph.D.

Northern Arizona University (NAU) — **Flagstaff, AZ, USA**
M.S. Statistics, CGPA 4.00/4.00 (With Distinction) — *April 2021*

Kwame Nkrumah University of Science and Technology (KNUST) — **Kumasi, Ghana**
B.Sc. Statistics, CWA 80.34/100 (First Class Honors) — *June 2018*
Thesis: A Randomized Clinical Design on the Impact of a New Dietary Treatment on Weight Loss.
Advisor: Mr. Isaac Akpor Adjei

RESEARCH INTEREST

- Distribution theory; Computational statistics; Time series analysis; Survival analysis; Fuzzy/ imprecise data; Multivariate data analysis;

WORK EXPERIENCE

Southern Methodist University (SMU) — **Dallas, TX, USA**
Lecturer/ Assistant Teaching Professor — *Aug. 2025 - Present*

University of Nevada Reno (UNR) — **Reno, NV, USA**
Graduate Teaching Assistant — *Aug. 2021 - Aug. 2025*

Northern Arizona University (NAU) — **Flagstaff, AZ, USA**
Graduate Teaching Assistant — *Aug. 2019 - April 2021*

Kwame Nkrumah University of Science and Technology (KNUST) — **Kumasi, Ghana**
Teaching Assistant — *Sept. 2018 - Aug. 2019*
National Service Personnel

HONORS & AWARDS

- **Graduate Student Travel Award** ($ 800), Department of Math & Stat, UNR — *May 2024*
- **Satisfactory Academic Progress Award (SAP)**, UNR — *May 2022, 2023 & 2024*
- **Valedictorian**, Best Graduating Student, College of Science (KNUST) — *Jul. 2018*
- **Best Graduating Student**, College of Science, College of Science Excellence Awards — *Jul. 2018*
- **Outstanding Best Male Student**, Faculty of Physical and Computational Sciences — *Jul. 2018*
- **Outstanding Best Student**, Association of Mathematics and Statistics Students — *May 2018*
- **Outstanding Best Student**, NUGS – KNUST, NUGS Excellence Awards — *Apr. 2018*

PUBLICATIONS

- **Saah, D.K.** and Kozubowski, T.J. (2025). A new class of extended Laplace distributions with applications to modeling contaminated Laplace data. *Journal of Computational and Applied Mathematics*, 465, 116588

- **Saah, D.K.** and Kozubowski, T.J. (2024). A new class of generalized distributions with links to density estimation, truncation, and imprecise data modeling [in preparation].
- **Saah, D.K.** and Kozubowski, T.J. (2024). New generalized distributions for modeling asymmetric and imprecise data [in preparation].
- **Saah, D.K.** and Kozubowski, T.J. (2024). Modeling multivariate imprecise data via new copula-driven generalized distributions [in preparation].

CONFERENCES & PRESENTATIONS (presenter in bf)

- 2nd Annual Data Science Conference, Nevada INBRE, University of Nevada, Reno, February 18-20, 2025, Participant & Workshop Attendee
- Invited Presentation "A new class of extended Laplace distributions with applications to modeling contaminated Laplace data" at the Applied Statistics and Data Science mini-symposium within the 9th European Seminar on Computing (ESCO 2024), June 10-14, 2024, Pilsen, Czech Republic [virtual presentation] (**Saah, D.K.** and Kozubowski, T.J.).
- Special Open Invited Lecture "A new class of extended Laplace distributions with applications to modeling contaminated Laplace data" at the 34th Conference on Mathematical Geophysics (CMG 2024) of the International Union of Geodesy and Geophysics held in the Indian Institute of Technology Bombay, June 2-7, 2024, Mumbai, India [virtual presentation in memory of the late Professor Illya Zaliapin] (**Saah, D.K.** and Kozubowski, T.J.).
- 1st Annual Data Science Conference, Nevada INBRE, University of Nevada, Reno, March 7–8, 2024, Participant & Workshop Attendee

TEACHING PHILOSOPHY

I am committed to fostering an inclusive and engaging learning environment that promotes active learning, critical thinking, and real-world application. My teaching approach integrates interactive problem-solving, hands-on data analysis, and collaborative group work to enhance student engagement. Recognizing diverse learning styles, I employ multiple instructional methods, including visual aids, interactive technology, and open discussions. I emphasize a growth mindset, providing constructive feedback and support to help students develop resilience and confidence in mathematics and statistics. My goal is to inspire students to appreciate the relevance of mathematical concepts and equip them with problem-solving skills applicable beyond the classroom.

TEACHING EXPERIENCE

Independent Course Instructor...

- **Statistical Methods for Engineers and Applied Scientists**, STAT/CS 4340 & OREM 3340, SMU
  - Taught three sections across different formats: an 8-day intensive (6 hrs/day, 17 students, Jan 2026), a standard semester (3 $\times$ 50 min/week, 64 students, Spring 2026), and an 11-day intensive (4 hrs/day, 8 students, May 2026)
  - Taught core probability and statistics for engineering applications: probability rules, discrete and continuous random variables, common distributions, sampling distributions, and the Central Limit Theorem
  - Covered statistical inference including point and interval estimation and one-sample hypothesis testing, with emphasis on applied problem-solving and interpretation
  - Developed all course assessments (exams, quizzes, homework assignments, and study materials); evaluated student performance, assigned final grades, and held regular office hours
- **Introduction to Statistical Methods**, STAT 2331, SMU

- Taught 3 sections per semester across Fall 2025 (270 students total) and Spring 2026 (230 students total); 3 sessions per week, 50 minutes each
- Developed all course assessments (4 exams, labs, and study guides); assigned final grades and held weekly office hours
- Taught a non-calculus-based introduction to statistical reasoning, including descriptive statistics, data visualization, and summary measures
- Introduced elementary probability, confidence intervals, hypothesis testing, and simple linear regression with emphasis on interpretation
- Integrated Excel for data analysis, visualization, and implementation of statistical methods

• **Precalculus I Expanded**, MATH 126.EE, UNR, Fall 2024, 42 students
  - Lectured 4 sessions per week (75 minutes each); created 6 exams, including finals, quizzes, and study guides; assigned final grades and held weekly office hours
  - Taught core algebraic techniques, including multiplying, dividing, and factoring polynomial expressions, solving polynomial and rational equations, and working with exponents and radicals.
  - Taught fundamental algebraic concepts, including polynomial, rational, exponential, and logarithmic functions; their graphs and applications; complex numbers; absolute value and quadratic inequalities; and systems of linear equations

• **Calculus I**, MATH 181, UNR, Summer 2024 & Winter 2025, 16 & 18 students, respectively
  - Lectured 5 sessions per week for summer, 6 sessions per week for winter (100 minutes each for summer, 200 minutes each for winter); created 4 exams, including finals, quizzes, and study guides; assigned final grades and held weekly office hours
  - Taught fundamental concepts of analytic geometry and calculus, including functions, graphs, limits, derivatives, and integrals.

• **Precalculus II**, MATH 127, UNR, Winter 2024 & 2023, 38 & 33 students, respectively
  - Lectured 5 sessions per week (3 hours each); created 4 exams, including finals, quizzes, and study guides; assigned final grades and held weekly office hours
  - Taught advanced precalculus topics, including trigonometric functions, identities, and equations; conic sections; complex numbers; polar coordinates; vectors; and systems of linear equations.

• **Fundamentals of College Mathematics**, MATH 120, UNR, Fall 2023, 40 students
  - Lectured 2 sessions per week (75 minutes each); created 4 exams, including finals, quizzes, and study guides; assigned final grades and held weekly office hours
  - Taught foundational mathematical concepts, including sets; probability and statistics; consumer mathematics; variation; geometry, and trigonometry for measurement; covered linear, quadratic, exponential, and logarithmic functions with an emphasis on problem-solving and real-world applications.

• **Precalculus I**, MATH 126, UNR, Summer 2022, 23 students
  - Lectured 5 sessions per week (95 minutes each); created 4 exams, including finals, quizzes, and study guides; assigned final grades and held weekly office hours
  - Taught fundamental algebraic concepts, including polynomial, rational, exponential, and logarithmic functions; their graphs and applications; complex numbers; absolute value and quadratic inequalities; and systems of linear equations

• **Quantitative Reasoning**, MAT 114, NAU, Summer 2021 & Spring 2020, 10 & 67 students, respectively
  - Lectured 4 sessions per week for summer (135 minutes each) and lectured 2 sessions per week (75 minutes each) for spring; created 4 exams, including finals, quizzes, and study guides; assigned final grades and held weekly office hours
  - Taught contemporary quantitative methods, including descriptive statistics, elementary probability, statistical inference, linear and exponential models of growth and decay, and discrete

models with practical applications.
  - Dedicated 5 hours per week to proctoring exams in the testing room and providing one-on-one tutoring to students in the open math lab at the Lumberjack Mathematics Center, supporting various mathematics courses and fostering student success.
- **Applied Statistics**, STA 270, NAU, Fall 2020 & Spring 2021, 80 & 74 students, respectively
  - Lectured 4 sessions per week (75 minutes each); created 4 exams, including finals, quizzes, and study guides; assigned final grades and held weekly office hours
  - Taught foundational statistical concepts, including graphical and quantitative data descriptions, binomial, normal, and t distributions; conducted instruction on one- and two-sample hypothesis tests, confidence intervals, simple linear regression, and correlation.
  - Participated in weekly coordination meetings to enhance and refine student learning.
- **Algebra for Precalculus**, MAT 108, NAU, Fall 2019, 115 students
  - Co-lectured 2 sessions per week (75 minutes each); created 4 exams, including finals, quizzes, and study guides; assigned final grades and held weekly office hours.
  - Taught algebraic operations, simplifying expressions, and analyzing functions; covered linear, absolute value, quadratic, cubic, and square root functions; solving equations and inequalities, and systems of equations through applied problem-solving techniques.
  - Dedicated 5 hours per week to proctoring exams in the testing room and providing one-on-one tutoring to students in the open math lab at the Lumberjack Mathematics Center, supporting various mathematics courses and fostering student success.

Recitation Course Instructor

- **Calculus I**, MATH 181, UNR, Spring 2024 & 2022, 50 & 56 students, respectively
  - Led a weekly recitation sessions (4 per week); graded quizzes, proctored exams, and held weekly office hours
- **Precalculus II**, MATH 127, UNR, Spring 2023 & 2025, 105 & 120 students, respectively
  - Led weekly recitation sessions (4 per week); graded quizzes, proctored exams, and held weekly office hours
- **Probability and Statistics**, STAT 352, UNR, Fall 2022,187 students
  - Led weekly recitation sessions (6 per week); graded quizzes, proctored exams, and held weekly office hours
- **Precalculus I**, MATH 126, UNR, Fall 2021, 111 students
  - Led weekly recitation sessions (6 per week); graded quizzes, proctored exams, and held weekly office hours
- **Assisted** with the instruction of Bayesian statistics, Time Series Methods and Analysis, as well as Survival Analysis at KNUST throughout my national service.
  Supervisor: Professor Nana Kena Frempong
  - Collaborated with my professor to design and execute lessons in alignment with the school's curriculum, aims, objectives, and philosophies.
  - Conducted statistical data analysis for researchers utilizing R and SPSS.
  - Performed administrative duties such as tracking attendance, grading assignments, maintaining student records, and creating teaching materials.

PROFESSIONAL DEVELOPMENTS & WORKSHOPS

Center for Teaching Excellence (CTE) — **SMU, USA**
33rd Annual Teaching Effectiveness Symposium — *Fall 2025*

- Attended the CTE Teaching Effectiveness Symposium keynote by Dr. David Yeager on motivating students through respect, mentorship, and purpose.
- Engaged in evidence-based strategies to improve student engagement and learning outcomes in

higher education.
- Reflected on pedagogical practices informed by research from the book *10 to 25*.

DCII / Office of Faculty Success **SMU, USA**
New Faculty Springboard: Awards & Research *Fall 2025*

- Participated in a professional development session that introduced faculty awards, research opportunities, and institutional support resources.
- Learned about the pathways for faculty recognition and funding within SMU.
- Networked with new faculty and representatives from the Office of Faculty Success.

UNR Nevada INBRE **UNR, USA**
Introduction to Version Control and GitHub Workshop *Nov. 2024*

- Explored the fundamentals of setting up and navigating GitHub for version control and project management.
- Learned to create and manage repositories tailored for academic and collaborative projects.
- Gained hands-on experience in contributing to and managing shared repositories for seamless team collaboration.
- Developed skills to publish a personal website using GitHub Pages to showcase projects and portfolios.

UNR Nevada INBRE **UNR, USA**
Data Strategy: Solving Real-World Problems with Generative AI Workshop *Nov. 2024*

- Participated in a hands-on workshop focused on leveraging Amazon Athena, Bedrock, and QuickSight for advanced data analytics.
- Explored techniques for efficient data manipulation and visualization using Amazon Web Services (AWS) tools.

UNR GSA Professional Development Series **UNR, USA**
Writing Teaching Statement *Sept. 2023*

NAU Teaching Day with José Bowen **NAU, USA**
Teaching With Fire *Nov. 2020*

Developing a Teaching Statement. **NAU, USA**
GTA Series *Jan. 2020*

INTERNSHIPS

United Airline Inc. **Chicago IL, USA**
Intern – Statistics and Operations Research *May 2023 - Aug. 2023*

- Led data-driven initiatives aimed at enhancing passenger experience, with a focus on optimizing meal distribution processes.
- Designed and developed a Python algorithm to simulate passenger meal preferences and loadings, leading to better meal planning and reduced wastage.
- Conducted in-depth analysis on passenger preferences following the introduction of a new entree, providing actionable insights to the catering team.
- Utilized time series forecasting techniques to predict pre-order meal demands.

Narh-Bita Hospital **Tema, Ghana**
Intern – Data Entry Clerk *May 2015 - Aug. 2015*

- Utilized R software to analyze data on weight reduction to assess the efficacy of a new medication.
- Collected and accessed patients' files.

SOFTWARE SKILLS

- **Programming & Data Analysis:** Proficient in R, Python, Mathematica, JMP/SAS, and SPSS.
- **Python Libraries:** Familiar with Tensorflow, Numpy, Pandas, Scikit-learn, and Keras.
- **Office Software:** Skilled in Microsoft Office (Word, PowerPoint, Excel) and LaTeX.

ACADEMIC SERVICE & PROFESSIONAL SOCIETIES

Mathematical Association of America (MAA) **USA**
Member *Dec. 2024 - Present*

Phi Kappa Phi (KPK) **NAU, USA**
Member *Mar. 2021 - Present*

American Statistical Association (ASA) **NAU, USA**
Member *Aug. 2019 - Present*

Association of Mathematics and Statistics Students (AMSS) **KNUST, Ghana**
Organizing Secretary *Aug. 2016 - May 2017*

- Collaborated with executives to plan and execute seminars, symposiums, field trips, and sports events.
- Coordinated student attendance from various classes for association events.
- Managed logistics, including auditorium bookings and refreshment provisions for attendees.

REFERENCES

- **Dr. Tomasz J. Kozubowski**
  Professor, Department of Mathematics and Statistics
  University of Nevada, Reno
  Email: tkozubow@unr.edu
  Phone: +1 (775) 784-6643

- **Dr. Anna Panorska**
  Professor, Department of Mathematics and Statistics
  University of Nevada, Reno
  Email: ania@unr.edu
  Phone: +1 (775) 784-6548

- **Dr. Andrey Sarantsev**
  Associate Professor, Department of Mathematics and Statistics
  University of Nevada, Reno
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  Phone: +1 (775) 784-6788